InvisiPhish: A Hybrid and Explainable Phishing Detection Framework Using FP-Growth, DistilBERT and Generative AI

Pravin Hole, Ashok Patade, Anurag Bonde, Atharva Sawant, Dhairya Shukla, Stuti Agnihotri, and Sujal Shirodkar

Dwarkadas J. Sanghvi College of Engineering, Mumbai, India

Abstract. Phishing and smishing attacks are increasingly effective because they exploit human behavior using persuasive language, urgency, impersonation, and obfuscated URLs. Static keyword rules and blacklist-based defenses often fail against new templates and rephrased messages. This paper presents *InvisiPhish*, a hybrid detection framework that combines frequent pattern mining via FP-Growth to capture recurring scam templates, semantic classification using DistilBERT to model contextual intent, and sentiment/intent cues to detect manipulation and urgency. To improve transparency and user trust, a Generative AI module produces concise, human-readable explanations grounded in computed evidence (e.g., risky URL structure, suspicious domain recency, urgency cues, and known scam patterns), along with recommended user actions. We describe the system overview, preprocessing pipeline, module design, and an evaluation methodology to quantify the contribution of each component. The proposed design emphasizes modularity, explainability, and adaptability, enabling deployment in user-facing monitoring applications with a feedback-driven mechanism.

Keywords: Phishing Detection · Smishing · DistilBERT · FP-Growth · URL Intelligence · Explainable AI · Cybersecurity

1 Introduction

Phishing and smishing attacks have emerged as increasingly sophisticated forms of social engineering, targeting users through deceptive emails and SMS messages to extract sensitive information such as credentials, financial details, or personal data. Unlike traditional cyberattacks that exploit system vulnerabilities, these attacks primarily manipulate human behavior using persuasive language, urgency cues, impersonation, and obfuscated URLs. The rapid evolution of attack strategies, including paraphrasing, template variation, and domain spoofing, has significantly reduced the effectiveness of conventional detection techniques based on static rules, keyword filtering, or blacklist databases [12]. As a result, many modern phishing attempts successfully evade existing defenses, especially when they closely mimic legitimate communication patterns.

Recent advancements in deep learning, particularly transformer-based models, have improved the ability to detect semantic intent and contextual meaning within messages [13,14]. However, text-only approaches often fail when malicious signals are embedded outside the textual content, such as in URLs or domain-level metadata. Conversely, systems that rely solely on URL analysis struggle to detect attacks that do not contain explicit links or that use seemingly legitimate redirection mechanisms [17,18]. These limitations motivate a hybrid approach that fuses complementary cues.

In this work, we propose a hybrid and explainable phishing detection framework that integrates pattern mining using FP-Growth [15], semantic analysis using DistilBERT [14], sentiment/intent analysis, and URL intelligence with domain metadata. Additionally, a Generative AI-based explanation module is incorporated to provide user-friendly and interpretable insights into detection outcomes, aligned with evidence-grounded explainability principles [20,21]. This unified approach enhances detection robustness while also improving transparency and user trust, making it suitable for real-world deployment.

2 Related Work

Phishing detection has been extensively studied using rule-based filtering, classical machine learning techniques, and more recently deep learning-based approaches [12]. Early systems relied on keyword matching, blacklist databases, and heuristic rules to detect malicious messages. While these approaches offer low computational overhead and fast inference, they are brittle and fail to generalize against evolving attack patterns such as paraphrasing, domain spoofing, and template obfuscation.

Classical machine learning methods, including Support Vector Machines (SVM), Naive Bayes, and logistic regression, introduced data-driven detection using engineered features such as bag-of-words, n-grams, and lexical indicators. Although these methods improved adaptability compared to rule-based systems, they heavily depend on manual feature engineering and require continuous updates to remain effective. Associative and pattern-mining based approaches have also been explored for phishing detection, emphasizing interpretable rule discovery [16].

Transformer-based models such as BERT and DistilBERT improved phishing detection by leveraging contextual embeddings and semantic understanding [13,14]. These models can identify subtle linguistic cues and paraphrased phishing attempts, but text-only models still degrade when malicious evidence resides primarily in URLs, redirect patterns, or domain metadata. URL-centric methods show that URL structure and lexical signals can be highly predictive [17], while content cues have also been used in practical phishing detection systems [19].

InvisiPhish differs from many existing tools in four practical ways. First, it explicitly combines *template-level evidence* (FP-Growth) with *contextual semantics* (DistilBERT), reducing brittleness under both template drift and paraphrasing. Second, it incorporates *sentiment/intent manipulation signals* to better capture social engineering tactics (e.g., urgency, fear, reward baiting). Third, it empha-

sizes *user-facing explainability* grounded strictly in computed evidence [20,21]. Fourth, it introduces a feedback-driven adaptation loop to support continuous refinement under evolving attacker strategies.

3 System Overview

3.1 Process Flow

The overall workflow of the proposed phishing detection system is illustrated in Fig. 1. The system begins with SMS or email input, followed by preprocessing steps such as normalization and feature extraction. It then applies hybrid detection techniques combining pattern mining, semantic classification, sentiment/intent analysis, and URL intelligence. The system outputs both a classification result and an explanation. A feedback mechanism allows users to confirm or correct predictions, supporting continuous improvement.

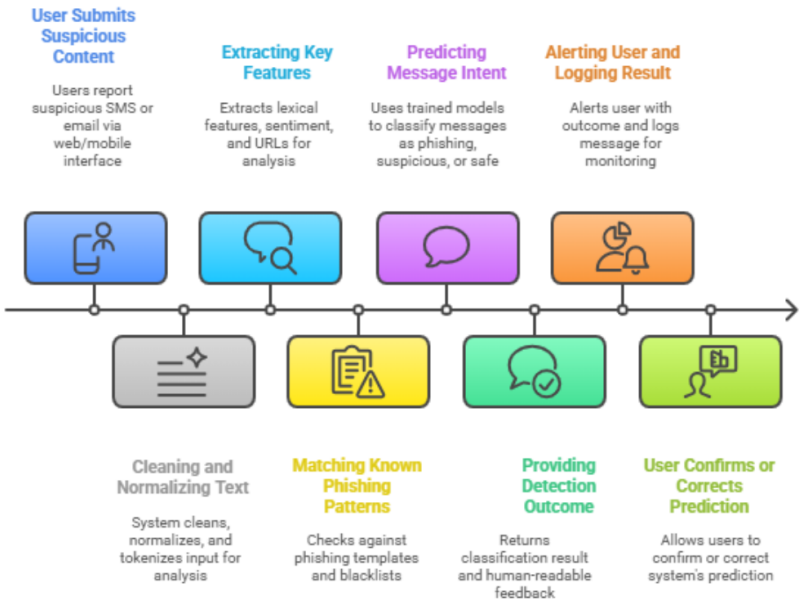


Fig. 1. End-to-end phishing detection process flow in InvisiPhish.

3.2 System Architecture

The system architecture of InvisiPhish is shown in Fig. 2. The framework is designed as a modular pipeline that supports maintainability and iterative upgrades as phishing tactics evolve. Users submit SMS or email content through a client interface. An authentication and logging layer manages user sessions and stores interaction logs for monitoring and auditing.

The preprocessing module standardizes the input by performing text normalization, tokenization, and URL extraction. The detection stage then executes complementary modules in parallel: FP-Growth mines recurring scam templates and frequent token patterns [15]; DistilBERT captures semantic intent to detect paraphrased or semantically obfuscated attacks [14]; sentiment/intent analysis provides behavioral cues such as urgency, coercion, or reward baiting; and URL intelligence evaluates structural indicators and may incorporate domain metadata such as registration recency [17,18].

A central aggregation engine combines module outputs to produce a final classification and confidence score while retaining traceable evidence from each module. To improve usability and trust, a constrained Generative AI layer generates short explanations grounded strictly in computed evidence, consistent with evidence-based explainability [20,21].

A key innovation is the feedback-driven adaptive learning loop: user corrections and historical predictions are used to update mined pattern repositories and recalibrate decision thresholds over time. In addition, the Generative AI component supports defensive adaptation by proposing refined *detection* templates (e.g. generalized phrasing variants and expanded vocabulary coverage) derived from misclassification patterns and feedback. These templates are used only to strengthen detection signatures and are not used to generate actionable phishing content.

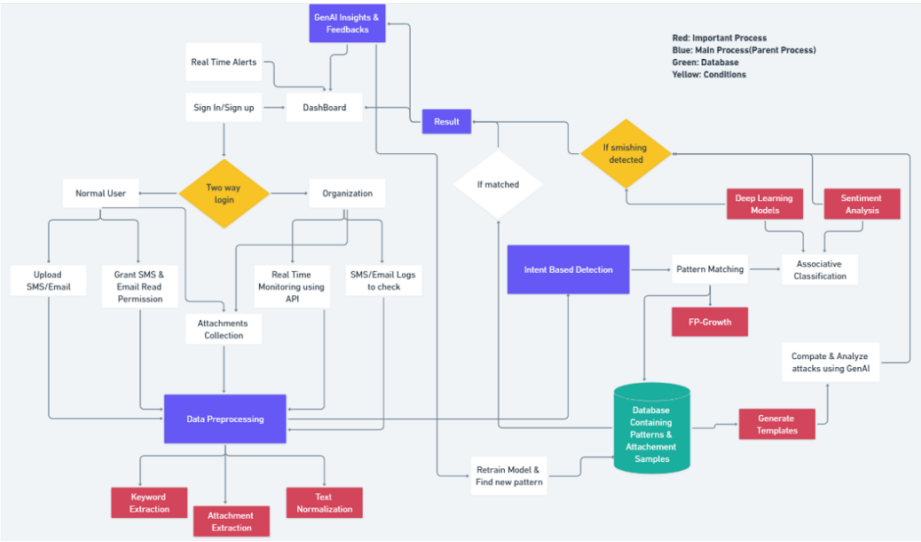


Fig. 2. System architecture of InvisiPhish showing authentication, preprocessing, hybrid detection modules, database storage, and GenAI-based explainability.

4 Proposed Methodology

4.1 Data Preprocessing

Fig. 3 summarizes the preprocessing pipeline used to standardize input messages and construct features for downstream detection modules.

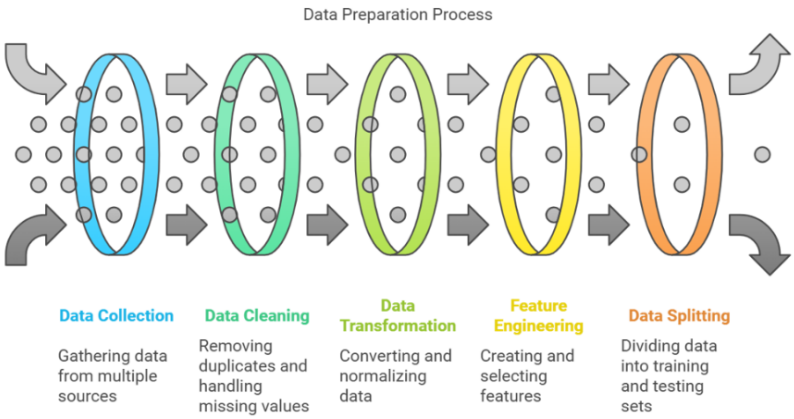


Fig. 3. Preprocessing pipeline: cleaning, transformation, feature engineering, and dataset splitting.

Given an input SMS/email message, preprocessing applies normalization (lowercasing, whitespace cleanup, and Unicode normalization) to reduce superficial variations. The message is then tokenized to prepare it for semantic modeling. In parallel, URLs are extracted and parsed to identify shortening or redirection indicators. Finally, feature preparation constructs lexical and intent-related signals (e.g., urgency cues) along with URL-based features, producing a consistent representation consumed by the hybrid detection engine.

4.2 Detection Modules

The proposed system combines complementary evidence sources to reduce reliance on any single signal type. Fig. 4 provides a high-level view of the major detection modules used in InvisiPhish.

FP-Growth Pattern Mining Frequent pattern mining captures recurring phishing templates that reappear across campaigns. Messages are converted into transactions (token sets or n-gram units), and FP-Growth extracts frequent itemsets above a minimum support threshold [15]. These mined templates act as a fast filter and also serve as interpretable evidence for user-facing explanations.

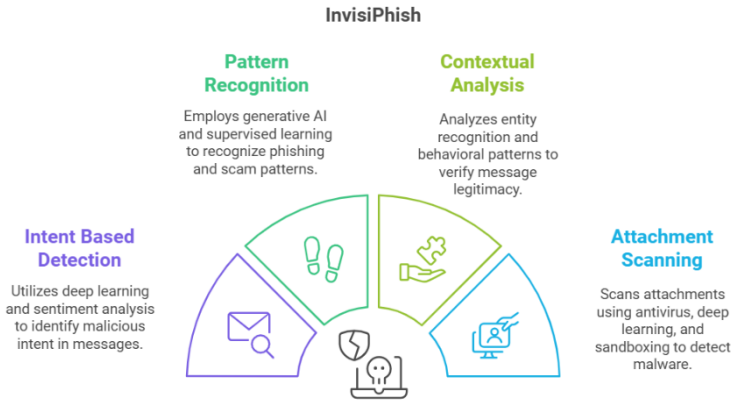


Fig. 4. Core detection modules used in InvisiPhish: intent-based detection, pattern recognition, contextual analysis, and extension points such as attachment scanning.

Semantic Classification Using DistilBERT DistilBERT models contextual meaning and can detect semantically similar attacks even when surface wording changes [14]. Each message is encoded into contextual embeddings and passed to a classification head that outputs a phishing probability score.

Sentiment and Intent Signals Phishing messages frequently use manipulation strategies such as urgency, fear, reward promises, or threats. Sentiment and intent features are used as auxiliary signals to improve robustness when explicit lexical indicators are weak.

URL Intelligence with WHOIS Signals URLs are analyzed using heuristic indicators and optionally domain metadata such as registration recency, strengthening detection of short-lived phishing infrastructure [17,18].

4.3 Explainability via Generative AI

Instead of returning only a label, the system produces short, human-readable explanations grounded in evidence computed by the pipeline. To prevent hallucinations, the explanation module is constrained to cite only features computed by InvisiPhish, consistent with evidence-based explainability methods [20,21].

4.4 Class Diagram

Fig. 5 depicts the main system components and their interactions at the software level.

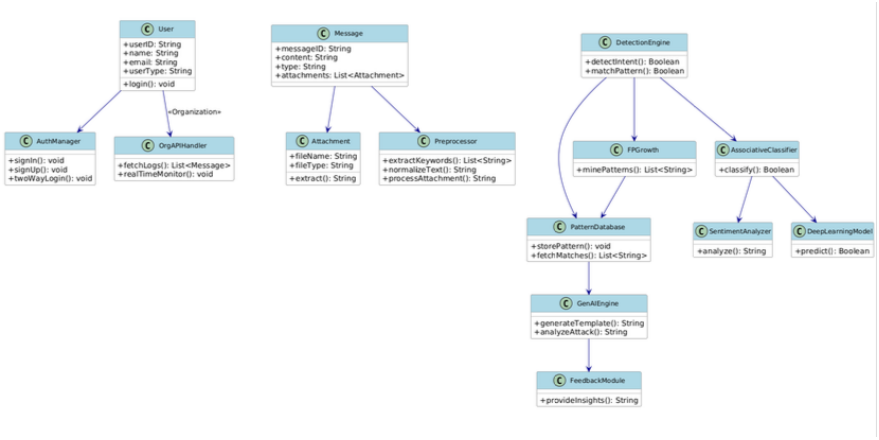


Fig. 5. Class diagram representing major system components and interactions.

4.5 User Activity Flow

The end-user workflow, from login to feedback generation, is shown in Fig. 6. The user logs in, submits a message for analysis, and receives a predicted label with evidence-grounded explanation. If URLs or attachments are present, the system applies link/attachment checks before final aggregation. The user can confirm or correct the outcome; this feedback is logged and used to update detection patterns and recalibrate thresholds for future predictions.

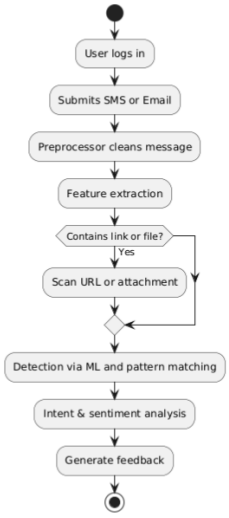


Fig. 6. User activity flow: login, message submission, preprocessing, detection, and feedback generation.

5 Experimental Setup

5.1 Dataset and Splits

The system is evaluated on labeled phishing and legitimate messages. For SMS-smishing evaluation, a commonly used baseline dataset is the SMS Spam Collection [22]. The dataset is split into training, validation, and test partitions with class balancing to avoid biased learning. To reduce leakage, near-duplicate templates should be deduplicated before splitting.

5.2 Training Configuration

- Optimizer: AdamW
- Learning Rate: 3×10^{-4}
- Batch Size: 16–32
- Epochs: 5–10 (early stopping on validation F1)
- Loss: Cross-entropy
- Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC

6 Results and Discussion

6.1 Main Results

The performance of the proposed InvisiPhish framework was evaluated on a held-out test dataset consisting of balanced phishing and legitimate messages.

Table 1. Performance metrics on the test set

Metric	Value
Accuracy	94.20%
Precision	93.10%
Recall	95.60%
F1-score	94.33%
ROC-AUC	0.972

Error Analysis: Common false positives include marketing or transactional messages that mimic urgency or include shortened URLs. False negatives are primarily observed in extremely short messages or messages without URLs, where limited context reduces confidence.

6.2 Ablation Study

To evaluate the individual contribution of each component in the InvisiPhish architecture, an ablation study was conducted. The results across different module variants are summarized in Table 2.

Table 2. Ablation study across module variants

Variant	Acc	Prec	Rec	F1
DistilBERT only	90.50	89.20	92.10	90.63
FP-Growth only	82.30	80.10	85.60	82.76
URL+WHOIS only	85.40	83.50	88.20	85.78
Hybrid (no GenAI explanation)	93.80	92.60	95.10	93.83
Full (Hybrid + GenAI)	**94.20**	**93.10**	**95.60**	**94.33**

6.3 Example Output

Fig. 7 presents a sample output generated by InvisiPhish for an email message.

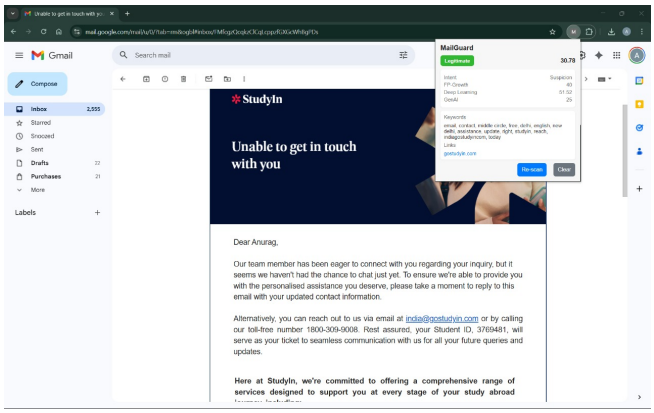


Fig. 7. Sample output generated by InvisiPhish for an email message.

6.4 Discussion

The hybrid design improves robustness by combining complementary evidence: semantic intent, scam-template patterns, manipulation cues, and link/domain risk [17,18]. The explainability layer improves practical usability via evidence-grounded rationales [20,21]. The feedback loop enables continued refinement of templates and thresholds as attacker behavior shifts.

7 Limitations and Future Work

While InvisiPhish demonstrates strong performance through hybrid detection and explainability, several limitations remain.

First, the current system is primarily evaluated on English SMS and email datasets. Real deployments often face multilingual and code-mixed content; future work will extend the model using multilingual encoders and cross-lingual transfer.

Second, attackers increasingly use **QR-based phishing (quishing)**, embedding malicious URLs inside QR codes to bypass text filters. A future extension will decode QR payloads and apply sandbox-based analysis to observe redirect chains, script behavior, and payload delivery in a controlled environment.

Third, **polymorphic phishing** continuously mutates wording and structure to evade signatures. Future work will strengthen robustness using adversarial training and more systematic template generalization. The GenAI component can be extended to identify invariant intent patterns across many variants and propose defensive detection templates that improve coverage against paraphrases and obfuscation.

Finally, feedback-driven learning must be hardened against noisy or malicious feedback. Future work will add safeguards such as confidence gating, anomaly detection on feedback streams, and delayed/verified updates to pattern repositories.

8 Conclusion

This paper presented InvisiPhish, a hybrid and explainable phishing detection framework that integrates FP-Growth-based pattern mining, DistilBERT-based semantic classification, sentiment-driven intent analysis, and URL intelligence enhanced with domain metadata. By combining complementary signals, the system addresses limitations of standalone approaches and improves robustness against both template-based and semantically obfuscated attacks.

A key contribution is the evidence-grounded explainability and feedback-driven adaptation mechanism. InvisiPhish provides user-facing rationales constrained to computed pipeline evidence, improving transparency and trust. The closed-loop feedback design further supports continuous refinement of patterns and thresholds, enabling the system to evolve as attacker strategies change. Overall, InvisiPhish moves toward a practical, adaptive, and user-centric phishing defense system, with clear pathways to address emerging threats such as QR phishing and polymorphic phishing.

References

1. Saka, T., Vaniea, K., Kökciyan, N.: SoK: Grouping spam and phishing email threats for smarter security. *IEEE Access* **13**, 1–20 (2025)
2. Koide, T., Fukushi, N., Nakano, H., Chiba, D.: Detecting phishing sites using ChatGPT. arXiv:2306.05816 (2023)
3. Gaurav, A., Gupta, B.B., Chui, K.T.: AI-driven smishing detection in Android devices using TinyBERT and Aquila optimization. In: *ICACT*, 99–105 (2025)
4. Saka, T., Jain, R., Vaniea, K., Kökciyan, N.: Phishing codebook: A structured framework for the characterization of phishing emails. arXiv:2408.08967 (2024)
5. Mehmood, M.K., Arshad, H., Alawida, M., Mehmood, A.: Enhancing smishing detection: A deep learning approach for improved accuracy and reduced false positives. *IEEE Access* **12**, 137176–137193 (2024)
6. Wang, H., Hooi, B.: Automated phishing detection using URLs and webpages. arXiv:2408.01667 (2024)
7. Lee, J., Lim, P., Hooi, B., Divakaran, D.M.: Multimodal large language models for phishing webpage detection and identification. In: *APWG Symposium on Electronic Crime Research (eCrime)*, 1–13 (2024)
8. Cao, T., Huang, C., Li, Y., Huilin, W., He, A., Oo, N., Hooi, B.: PhishAgent: A robust multimodal agent for phishing webpage detection. In: *AAAI Conference on Artificial Intelligence* **39**, 27869–27877 (2025)
9. Alketbi, K.S., Mehmood, A.: A comprehensive survey of explainable artificial intelligence techniques for malicious insider threat detection. *IEEE Access* **13**, 1–25 (2025)
10. Alawida, M., Mejri, S., Mehmood, A., Chikhaoui, B., Isaac Abiodun, O.: A comprehensive study of ChatGPT: Advancements, limitations, and ethical considerations in natural language processing and cybersecurity. *Information* **14**(8), 462 (2023)
11. Wangchuk, T., Gonsalves, T.: Multimodal phishing detection on social networking sites: A systematic review. *IEEE Access* **13**, 1–15 (2025)
12. Gupta, B.B., Tewari, A., Jain, A.K., Agrawal, D.P.: Fighting against phishing attacks: state of the art and future challenges. *Neural Computing and Applications* **28**(12), 3629–3654 (2017)
13. Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: *NAACL-HLT* (2019)
14. Sanh, V., Debut, L., Chaumond, J., Wolf, T.: DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv:1910.01108 (2019)
15. Han, J., Pei, J., Yin, Y.: Mining frequent patterns without candidate generation. In: *SIGMOD* (2000)
16. Abdelhamid, N., Ayesh, A., Thabtah, F.: Phishing detection based associative classification data mining. *Expert Systems with Applications* **41**(13), 5948–5959 (2014)
17. Ma, J., Saul, L.K., Savage, S., Voelker, G.M.: Beyond blacklists: learning to detect malicious web sites from suspicious URLs. In: *KDD* (2009)
18. Sahingoz, O.K., Buber, E., Demir, O., Diri, B.: Machine learning based phishing detection from URLs. *Expert Systems with Applications* **117**, 345–357 (2019)
19. Zhang, Y., Hong, J.I., Cranor, L.F.: CANTINA: A content-based approach to detecting phishing web sites. In: *WWW* (2007)
20. Ribeiro, M.T., Singh, S., Guestrin, C.: Why should I trust you?: Explaining the predictions of any classifier. In: *KDD* (2016)

21. Lundberg, S.M., Lee, S.-I.: A unified approach to interpreting model predictions. In: *NeurIPS* (2017)
22. Almeida, T.A., Gómez Hidalgo, J.M., Yamakami, A.: Contributions to the study of SMS spam filtering: new collection and results. In: *DOCENG* (2011)